

# Spatial ML - class 4 - Degree of Urbanisation (DEGURBA)

Total points 90/100 ?

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**1. Why does the DEGURBA methodology use population grids instead of administrative boundaries?** \*10/10

- ☒ Administrative boundaries do not always reflect the real spatial structure of settlements
- ☐ Population grids are easier to store in spatial databases
- ☐ Administrative units cannot contain population data properly
- ☐ Population grids eliminate the need for spatial projections



**2. In the DEGURBA Level 1 classification, what distinguishes an urban centre from an urban cluster?**

\*10/10

- ☐ Whether the grid cells belong to administrative cities
- ☒ Population density and total population thresholds
- ☐ Distance to the nearest capital city
- ☐ The number of surrounding municipalities

**3. Which spatial contiguity rule is used when identifying urban centres?**

10/10

- ☐ Queen contiguity
- ☐ Distance-based neighbours
- ☒ Rook contiguity
- ☐ K-nearest neighbours



**4. Which of the following datasets is NOT required to run the DEGURBA classification in the workflow used in class?** \*10/10

- ☐ Population grid
- ☐ Built-up surface grid
- ☒ Road network grid
- ☐ Land grid

**5. In the R workflow, why must the study area be transformed to the Mollweide projection before cropping GHSL rasters?** \*10/10

- ☐ Because raster data can only be processed in Mollweide
- ☒ Because the GHSL rasters are provided in this projection
- ☐ Because the flexurba package requires EPSG:4326
- ☐ Because spatial joins require identical projections



**6. What is the main purpose of the preprocessing step before running the DEGURBA classification?** \*10/10

- ☐ To merge administrative units with raster cells
- ☒ To ensure the raster layers share the same structure and calculate density
- ☐ To convert rasters to polygons for plotting
- ☐ To reduce the resolution of the population grid and calculate density

**7. In the DEGURBA Level 1 grid classification, which category corresponds to the value 3?** \*0/10

- ☐ Urban centre
- ☒ Rural grid cell
- ☐ Urban cluster
- ☐ Water grid cell



**8. Why are built-up land data included in the DEGURBA algorithm? \***

10/10

- ☐ To estimate commuting flows between cities
- ☐ To identify administrative boundaries and incorporate into the workflow
- ☒ To prevent infrastructure areas with low population from splitting urban clusters
- ☐ To measure accessibility to services

**9. What is the purpose of converting the DEGURBA raster into polygons (as.polygons()) in the script?**

\*10/10

- ☐ To reduce the number of raster cells
- ☒ To enable vector spatial operations such as spatial joins
- ☐ To convert the data to a different projection
- ☐ To simplify the classification thresholds



**10. When linking firms to DEGURBA classes using `st_join()`, what condition must be satisfied?** 10/10

- ☒ Both datasets must use the same coordinate reference system
- ☐ The firm dataset must contain population data
- ☐ Both datasets must be rasters
- ☐ The spatial join must use queen contiguity

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